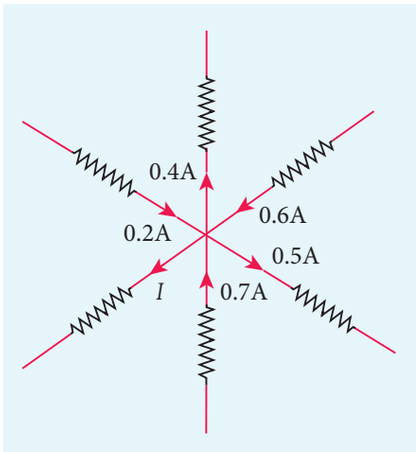
$$I_1 + I_2 - I_3 - I_4 - I_5 = 0$$

(or)

$$I_1 + I_2 = I_3 + I_4 + I_5$$

EXAMPLE 2.20

From the given circuit find the value of I.



Solution

Applying Kirchoff's rule to the point P in the circuit,

The arrows pointing towards P are positive and away from P are negative.

Therefore, 0.2A – 0.4A + 0.6A – 0.5A + 0.7A – I = 0

1.5A – 0.9A – I = 0

0.6A – I = 0

I = 0.6 A

2.5.2 Kirchhoff's Second rule (Voltage rule or Loop rule)

It states that in a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit. This rule follows from the law of conservation of energy for an isolated system (The energy supplied by the emf sources is equal to the sum of the energy delivered to all resistors).

The product of current and resistance is taken as positive when the direction of the current is followed. Suppose if the direction of current is opposite to the direction of the loop, then product of current and voltage across the resistor is negative. It is shown in Figure 2.24 (a) and (b). The emf is considered positive when proceeding from the negative to the positive terminal of the cell. It is shown in Figure 2.24 (c) and (d).

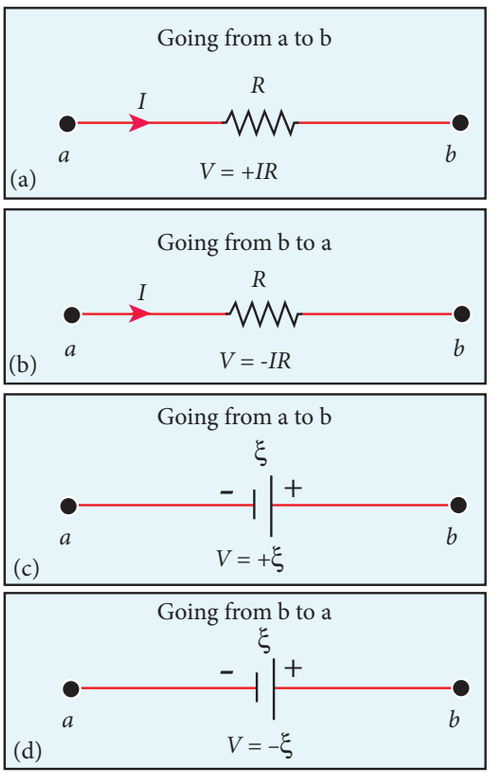


Figure 2.24 Kirchhoff voltage rule

Kirchhoff voltage rule has to be applied only when all currents in the circuit reach a steady state condition (the current in various branches are constant).

EXAMPLE 2.21

The following figure shows a complex network of conductors which can be divided into two closed loops like ACE and ABC. Apply Kirchoff's voltage rule.